1.1 What is GIS?

Geographic information systems

The concept of 'space' is part of much research in the humanities and in linguistics. Archaeological finds, historical events and linguistic phenomena, among others, are partly defined by their location and distribution. Because of this, these disciplines often need to structure, analyse and visualise information spatially. This can be done in different ways: analogue or digital, with general tools or with specialised tools.

Geographic information systems (GIS) belong to this last category. These (digital) systems were developed specifically to process **geo-information**, that is, information with a spatial dimension.

Geographic information systems offer their users a wide range of tools and functions that can relate to every step of the data processing workflow:

1. **Creating** digital geodata (manually or automatically)
2. **Structuring**, **modelling**, **storing** and **managing** spatial information
3. **Editing** both the spatial and the non-spatial information
4. **Querying** geographic data
5. **Visualising** and exporting maps

Not every GIS application covers all of the aspects above. There is a great variety of GIS software: some packages specialise in specific aspects (for example storing geodata or creating maps), others take a more integrated approach.
There are also large differences in user interaction: some applications have an advanced graphical user interface, others have none at all.
Finally, there are applications meant for local use (desktop software) as well as server-based systems. The latter offer extra possibilities (for example for collaboration), but they also require a lot of system administration work. For this reason, they are usually out of reach for individual researchers.

Desktop GIS

In this very diverse GIS landscape, ArcGIS and QGIS are among the most widely used software packages. Both offer a wide range of functions in an integrated, user-friendly graphical interface. Both can also easily be installed as a standalone application on the end user's computer. Because they are fairly easy to learn, easy to install and offer many possibilities, these packages have helped to make GIS popular in the humanities and in linguistics.

The **basic principles** of ArcGIS and QGIS are the same:

- **The map canvas of these GIS applications can be seen as a coordinate system.**
  On this coordinate system, (spatial) information can be positioned based on an x value and a y value (in a *geographic* coordinate system, these are the longitude and the latitude, expressed in degrees; see the chapter on coordinate systems). So the data we mainly work with in a GIS environment are data that **contain coordinates**. If they do not, a GIS will not be able to position the data (correctly) in space. By default, the axes and the coordinate grid itself are not visible in the map canvas of these geographic information systems. However, they are still present in the background.
- **Non-spatial information can be linked to spatial information.**
  For example, you can have a data object that represents a point with the coordinates 3.723° east longitude

and 51.045° north latitude (spatial information). About this point you can say that it is the *Book Tower* (name), that it was built in the period 1936–1942 (period) and that it is owned by Ghent University (owner). Name, construction period and owner are of course non-spatial data. Both the spatial and the non-spatial dimensions of a dataset can be queried and edited in these geographic information systems.
- **In these GIS environments, data are ideally stored in (one or more) thematic information layers.**
  Examples of such layers are roads (and waterways), buildings, boundaries, contour lines… In a GIS environment, different information layers can easily be combined, compared and processed.

Besides these general principles, ArcGIS and QGIS also have more similarities than differences in their user interface and functions. If you know one of these systems well, you will quickly learn the other one, and the other way round. Still, there are of course some (small and larger) differences in use, performance and possibilities. Neither package is better than the other in every respect: both have advantages and disadvantages.

ArcGIS (its current desktop application is called *ArcGIS Pro*) is a commercial product of the company Esri. To be able to use it, you therefore need to purchase a licence. QGIS, on the other hand, is *open source* software and completely free to use. For this reason, the practical parts of this learning path will always use QGIS.

QGIS

To follow this learning path properly, we recommend that you install QGIS on your own computer. You can easily download it from the project website https://www.qgis.org. Choose the installer for your operating system (Windows, macOS or Linux).

The download page offers two versions: the **Long Term Release (LTR)**, which is the most stable version, and the **latest release**, which has the newest functions.

GIS software evolves quickly, and new versions follow one another at a fast pace. This learning path is based on **QGIS 4.x** (the screenshots were made with QGIS 4.2.3).

> Tip
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> Changes to the functions and the interface are usually small, but even between minor versions (4.0, 4.2, …) differences occur in practice: for example, a button in a slightly different place or with a slightly different name. If your screen does not look exactly like a screenshot, look for a button or menu item with a similar name nearby.